

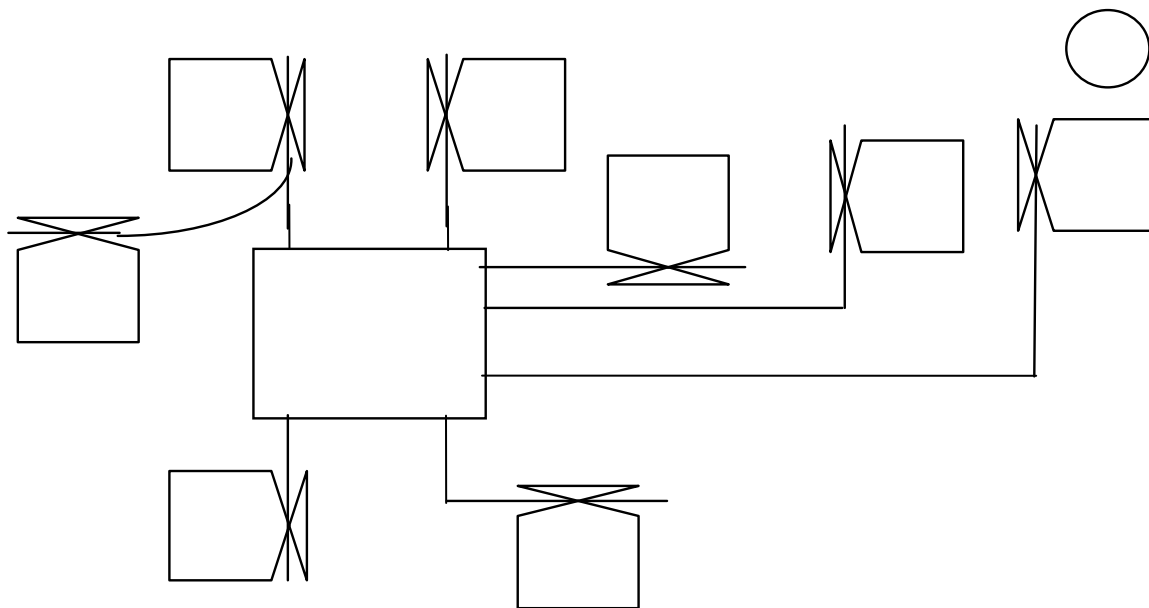
Additional examples of exam questions

Question 1

- 1) In connection with the Lintul model we have discussed water limited crop growth: the distribution of assimilates over the organs is changed if water limitation occurs. Discuss the basic model idea on *how* the distribution of assimilates will change if water shortage occurs. Consider in your explanation the changes in the distribution of assimilates (i) among roots and shoot, (ii) among the above ground plant parts and (iii) discuss the connection with the crop developmental stage.
- 2) In Lintul only one model layer is used to keep track of the soil water for the crop. Explain why this is a reasonable assumption.

Question 2

The figure below shows a part of the relational diagram of the water balance of the Lintul2 model, except for the names of the state, the flows, their directions and their feedbacks. The circle is an auxiliary (or help) variable to indicate the LAI.



- 1) Incorporate the names of the state and the rates in the relational diagram.
- 2) Indicate the directions of the flows in the relational diagram.
- 3) Incorporate the feedbacks into the relational diagram.
Clearly explain why a feedback occurs. If there is no feedback, also indicate why it is absent. By these explanations you in fact explicitly give the assumptions underlying this model on water dynamics.
Please note that you can only obtain points for this question if you also give explanations.
- 4) The auxiliary (or help) variable (indicated by the circle) is LAI. Indicate which rates are affected by LAI and explain why.
Please note that you can only obtain points for this question if you also give an explanation.
- 5) Explain the essential assumption of the Lintul2 water balance model.
Which characteristic moisture content data are needed for the tipping bucket model?